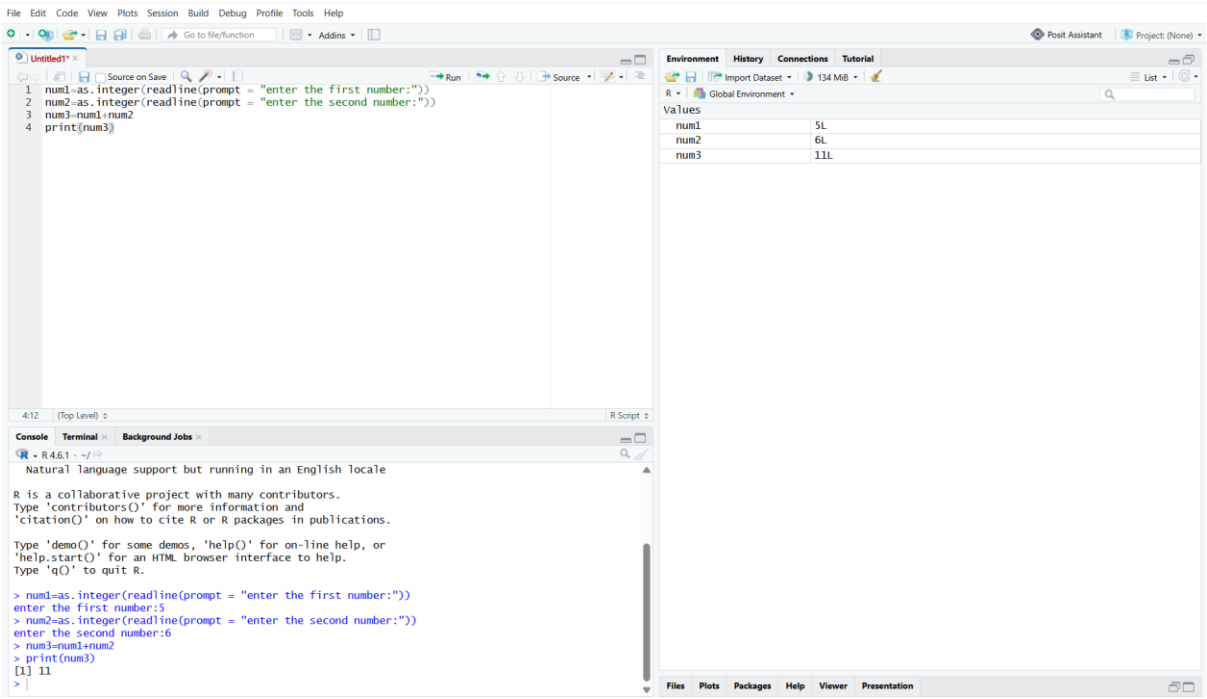
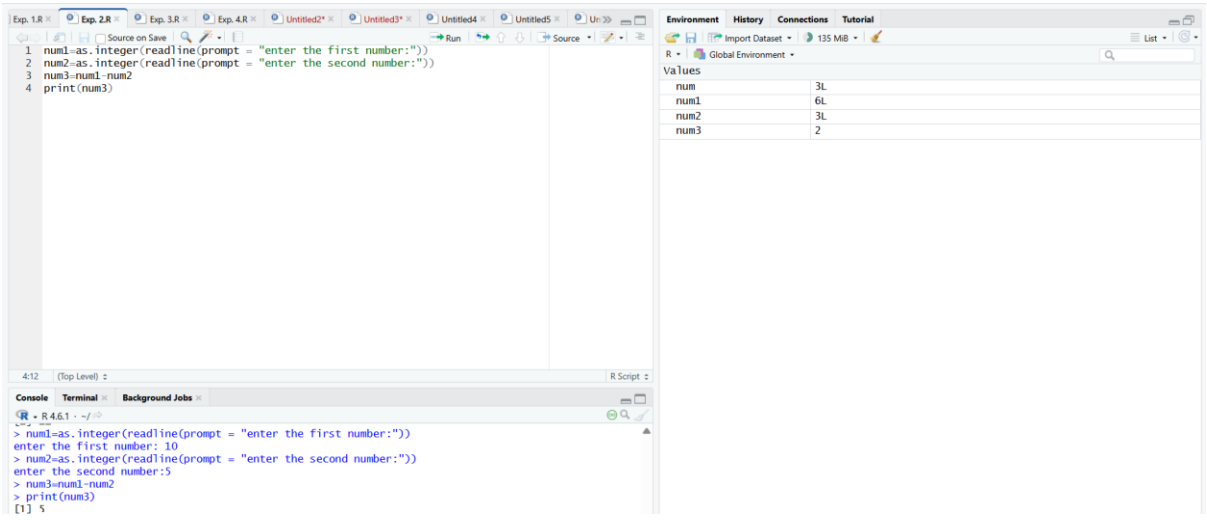
EXPERIMENTS 1 TO 20

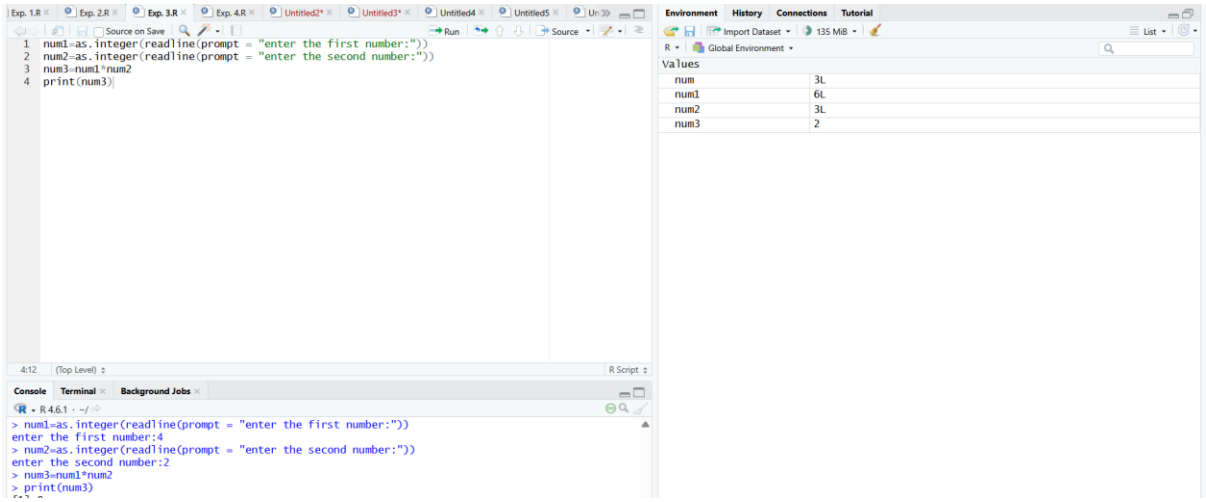
1.



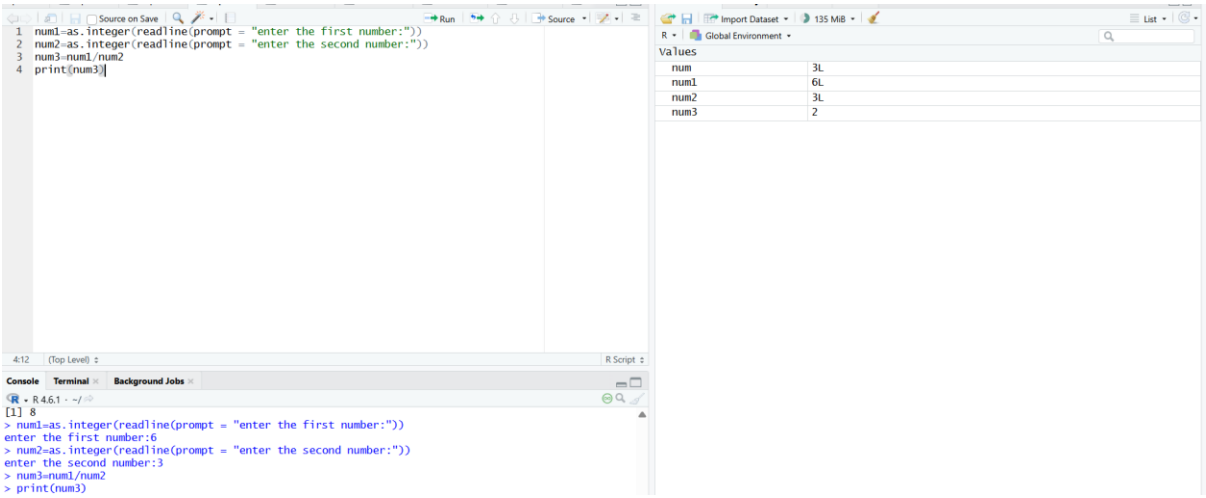
2.



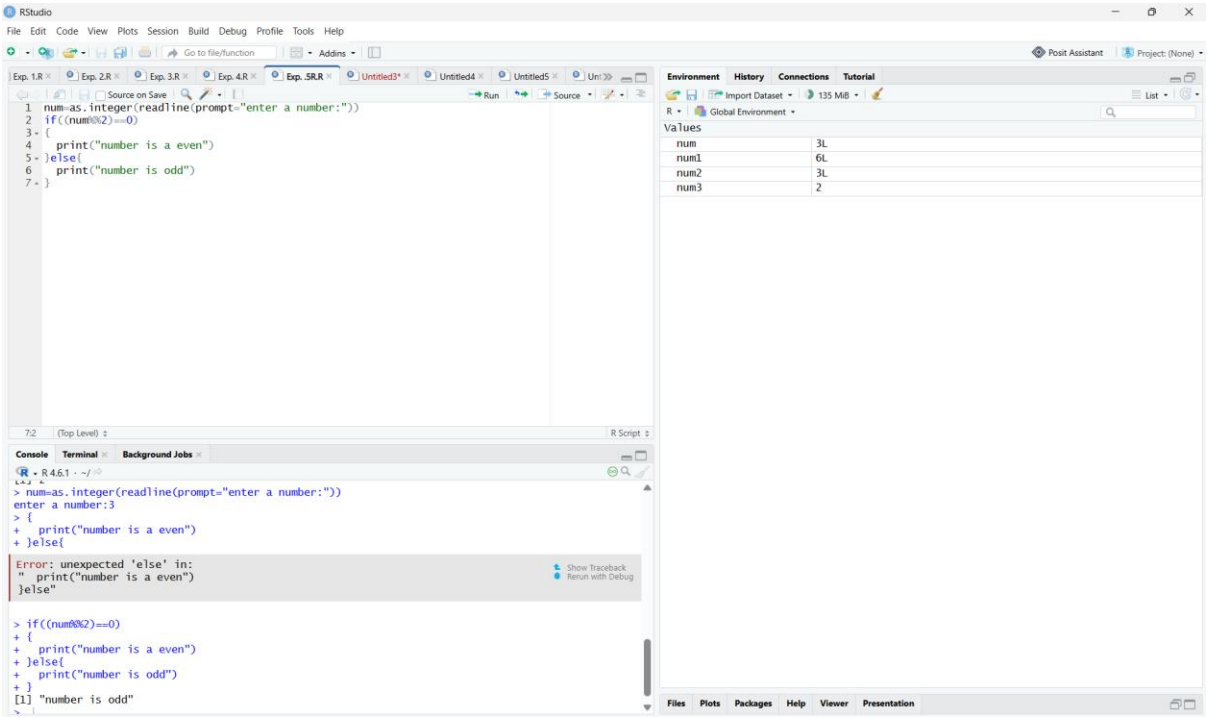
3.


num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1*num2
print(num3)
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:4
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:2
> num3=num1*num2
> print(num3)
Values
num 3L
num1 6L
num2 3L
num3 2

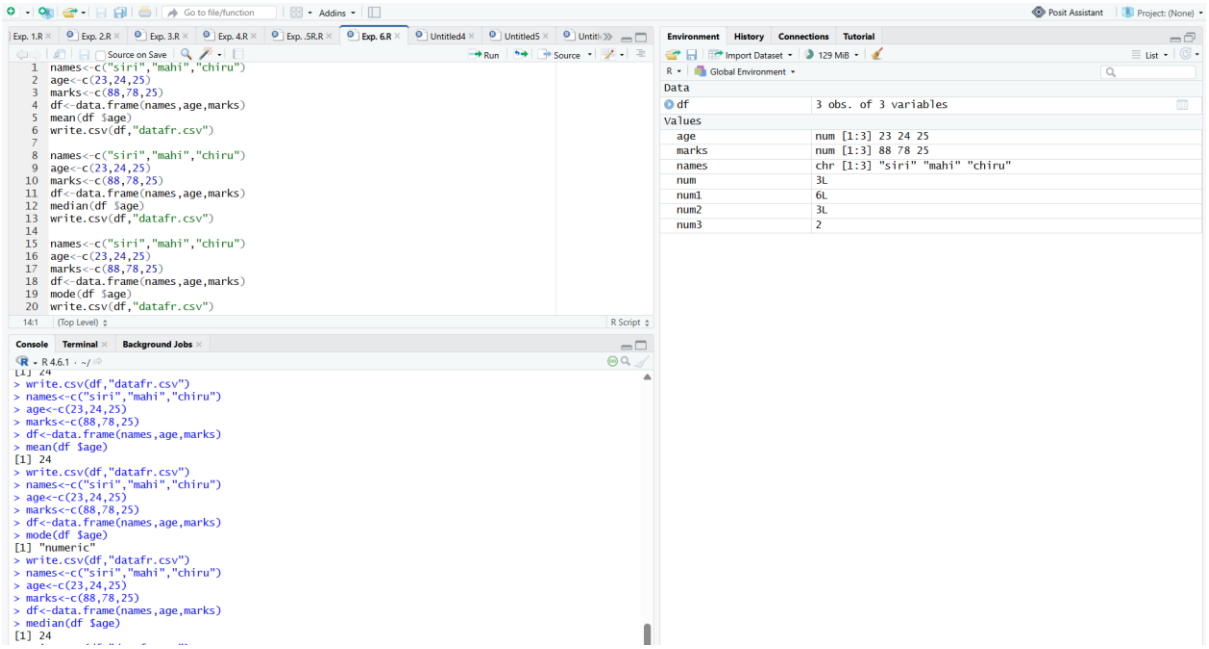
4.


num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1/num2
print(num3)
[1] 8
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:6
> num2=as.integer(readline(prompt = "enter the second number:"))
enter the second number:3
> num3=num1/num2
> print(num3)
Values
num 3L
num1 6L
num2 3L
num3 2

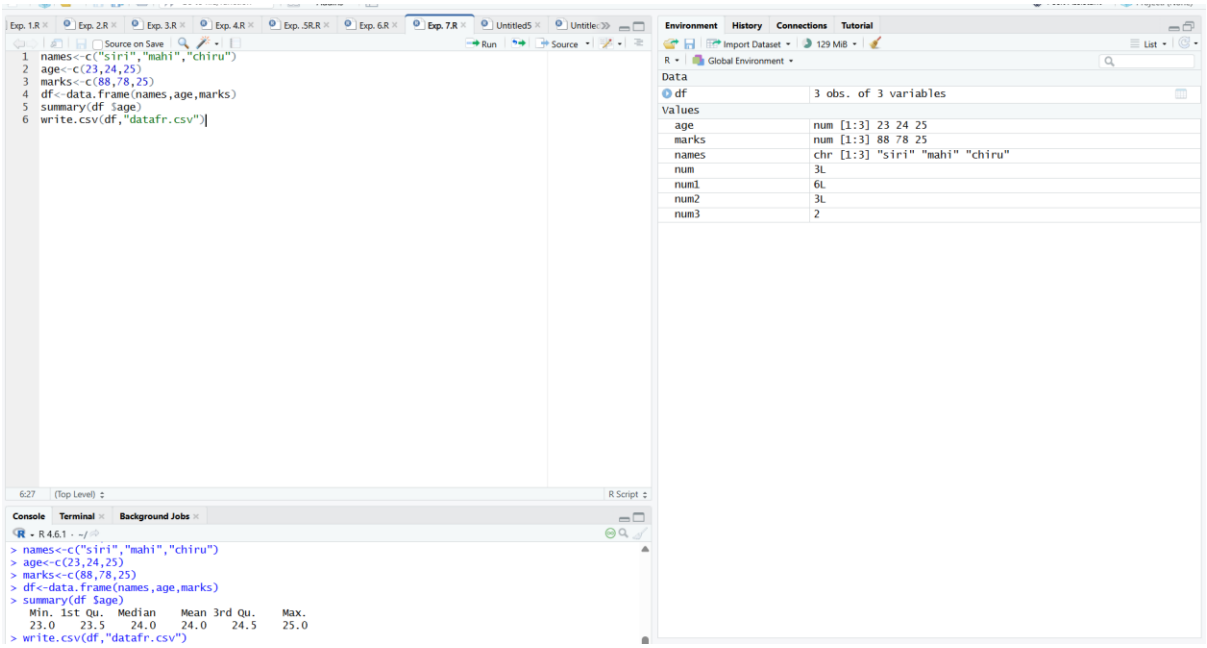
5.


num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
print("number is a even")
}else{
print("number is odd")
}
num=as.integer(readline(prompt="enter a number:"))
enter a number:3
{
print("number is a even")
}else{
Error: unexpected 'else' in:
" print("number is a even")
}else"
if((num%%2)==0)
{
print("number is a even")
}else{
print("number is odd")
}
[1] "number is odd"
num 3L
num1 6L
num2 3L
num3 2

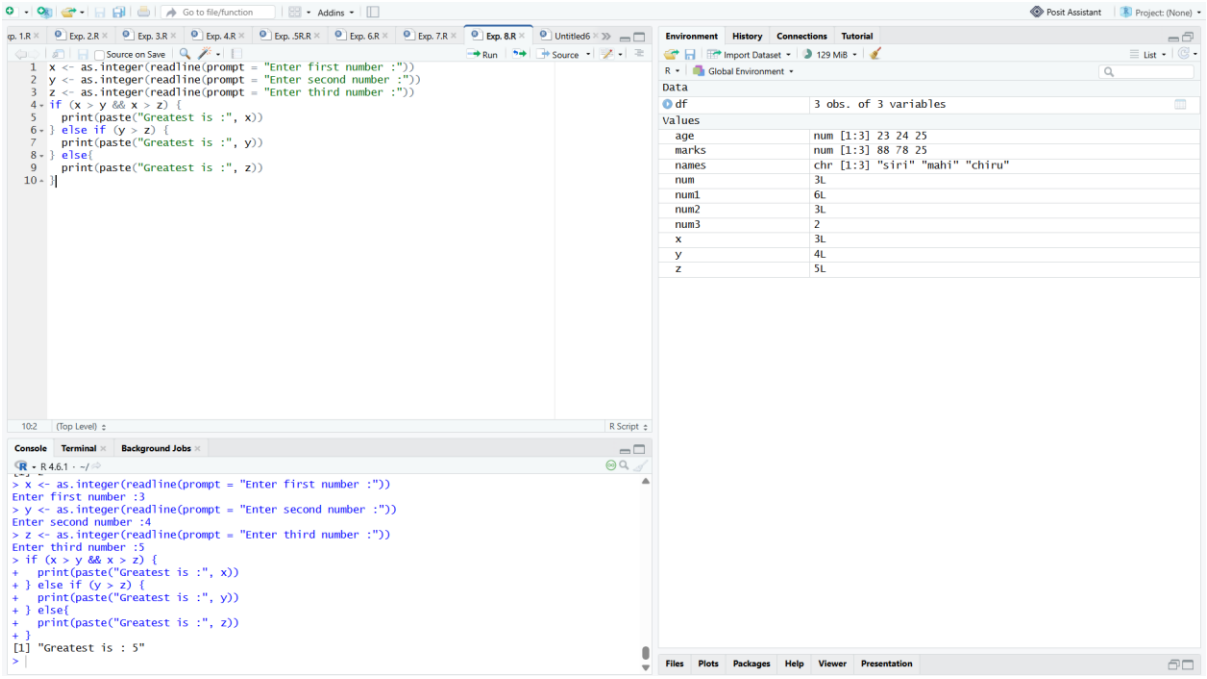
6.


names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
write.csv(df,"datafr.csv")
median(df $age)
mode(df $age)
> mean(df $age)
[1] 24
> median(df $age)
[1] 24
Data
df 3 obs. of 3 variables
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"

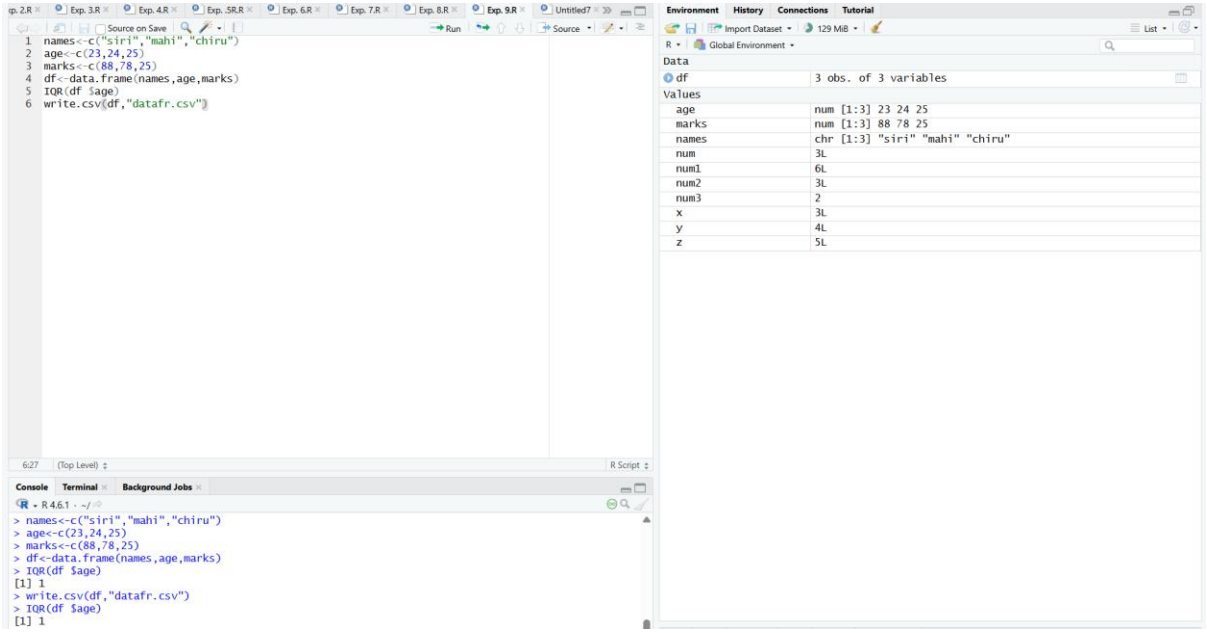
7.


names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
summary(df $age)
write.csv(df,"datafr.csv")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> summary(df $age)
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.0 23.5 24.0 24.0 24.5 25.0
> write.csv(df,"datafr.csv")
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 6L
num2 3L
num3 2

8.


x <- as.integer(readline(prompt = "Enter first number :"))
y <- as.integer(readline(prompt = "Enter second number :"))
z <- as.integer(readline(prompt = "Enter third number :"))
if (x > y && x > z) {
print(paste("Greatest is :", x))
} else if (y > z) {
print(paste("Greatest is :", y))
} else{
print(paste("Greatest is :", z))
}
> x <- as.integer(readline(prompt = "Enter first number :"))
Enter first number :3
> y <- as.integer(readline(prompt = "Enter second number :"))
Enter second number :4
> z <- as.integer(readline(prompt = "Enter third number :"))
Enter third number :5
> if (x > y && x > z) {
+ print(paste("Greatest is :", x))
+ } else if (y > z) {
+ print(paste("Greatest is :", y))
+ } else{
+ print(paste("Greatest is :", z))
+ }
[1] "Greatest is : 5"
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 3L
num1 6L
num2 3L
num3 2
x 3L
y 4L
z 5L

9.

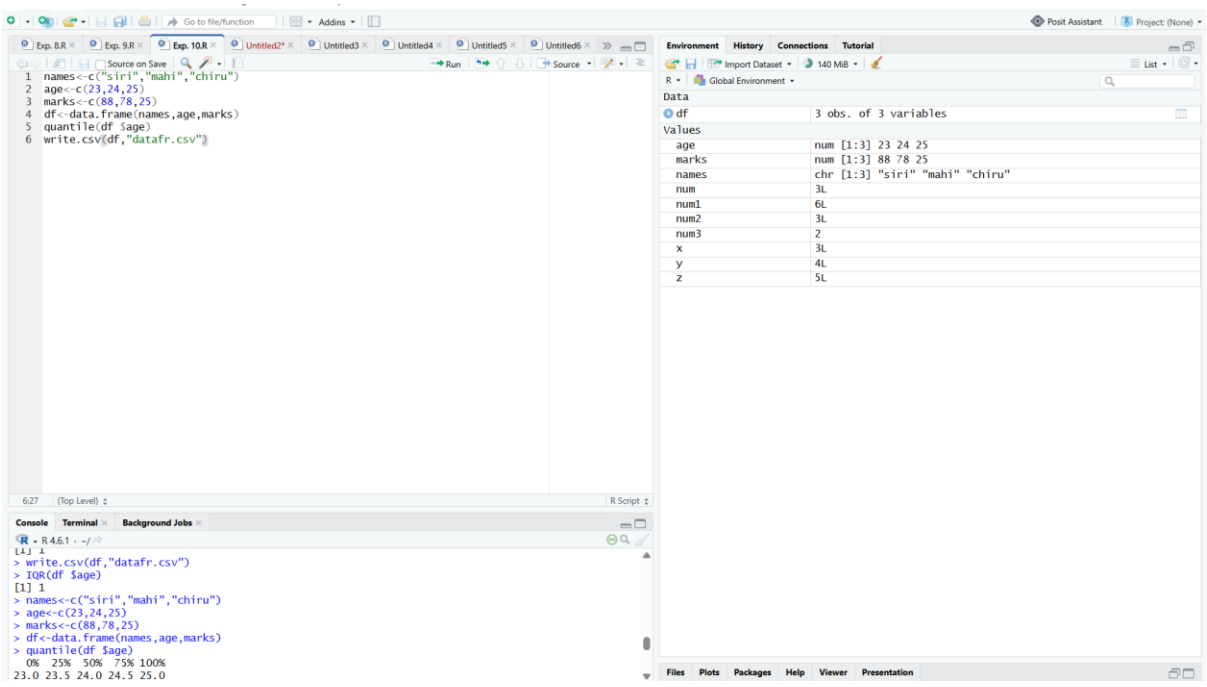


```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
write.csv(df,"datafr.csv")
```

```
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> IQR(df $age)
[1] 1
> write.csv(df,"datafr.csv")
> IQR(df $age)
[1] 1
```

Data	
df	3 obs. of 3 variables
Values	
age	num [1:3] 23 24 25
marks	num [1:3] 88 78 25
names	chr [1:3] "siri" "mahi" "chiru"
num	3L
num1	6L
num2	3L
num3	2
x	3L
y	4L
z	5L

10.

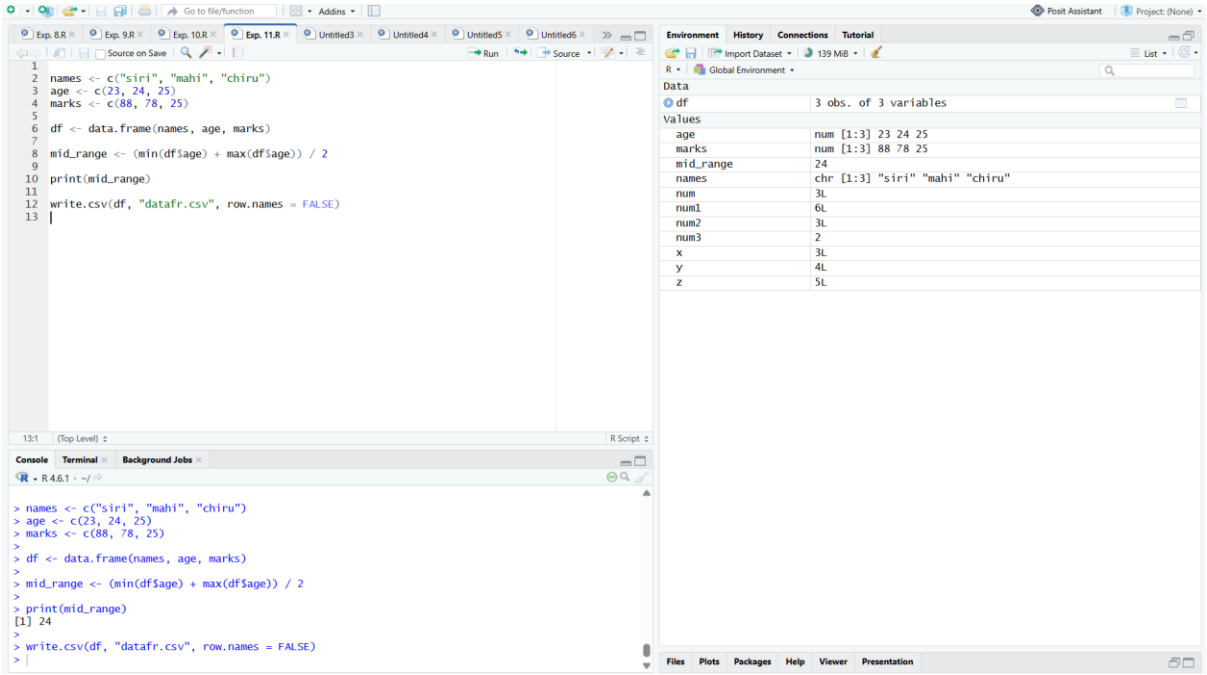


```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
write.csv(df,"datafr.csv")
```

```
> write.csv(df,"datafr.csv")
> IQR(df $age)
[1] 1
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> quantile(df $age)
  0%  25%  50%  75% 100%
23.0 23.5 24.0 24.5 25.0
```

Data	
df	3 obs. of 3 variables
Values	
age	num [1:3] 23 24 25
marks	num [1:3] 88 78 25
names	chr [1:3] "siri" "mahi" "chiru"
num	3L
num1	6L
num2	3L
num3	2
x	3L
y	4L
z	5L

11.



```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)

df <- data.frame(names, age, marks)

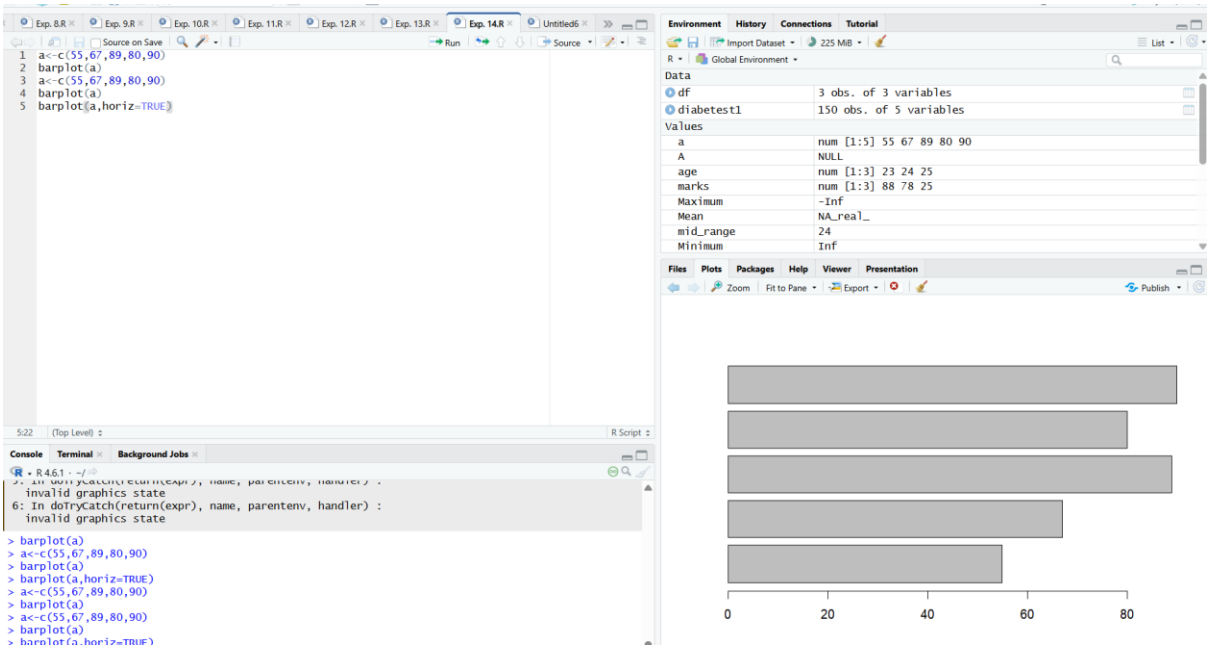
mid_range <- (min(df$age) + max(df$age)) / 2

print(mid_range)

write.csv(df, "datafr.csv", row.names = FALSE)
```

```
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
>
> df <- data.frame(names, age, marks)
>
> mid_range <- (min(df$age) + max(df$age)) / 2
>
> print(mid_range)
[1] 24
>
> write.csv(df, "datafr.csv", row.names = FALSE)
>
```

14.

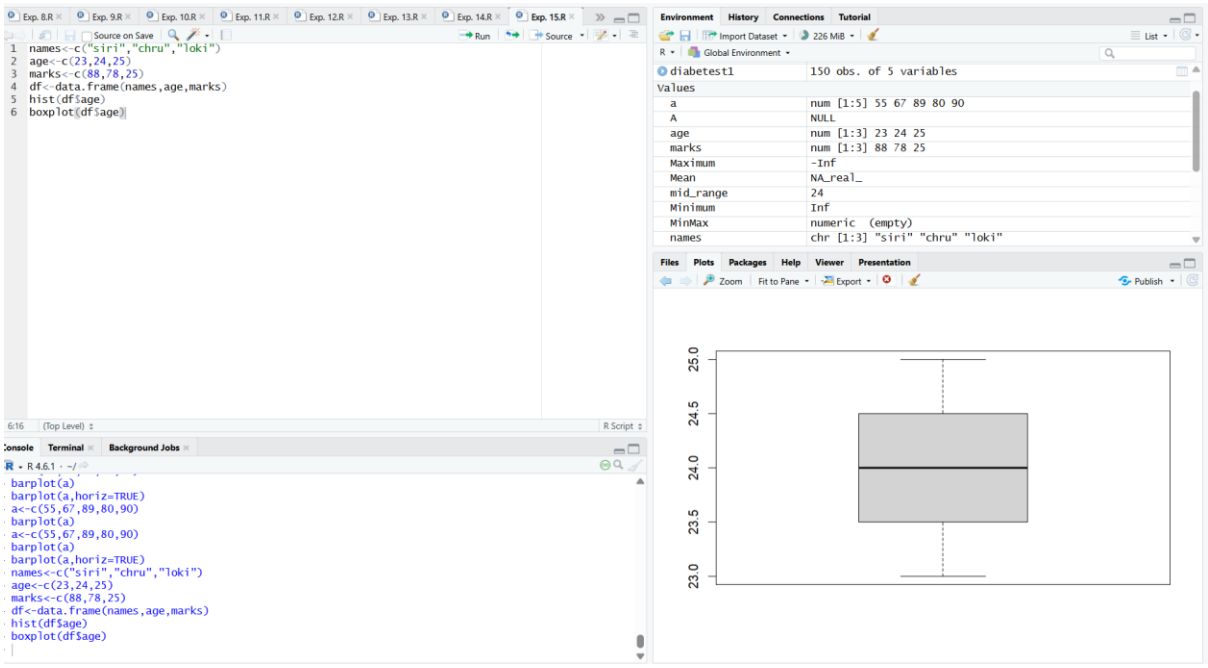


```
a<-c(55,67,89,80,90)
barplot(a)
a<-c(55,67,89,80,90)
barplot(a)
barplot(a,horiz=TRUE)
```

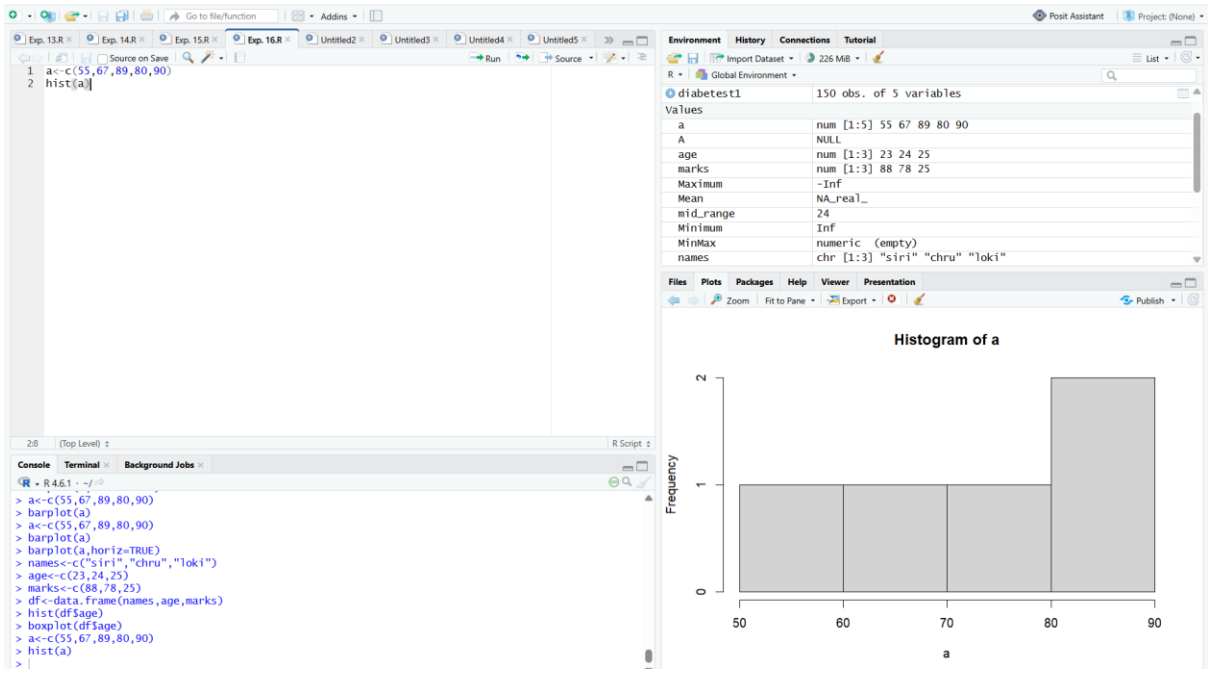
```
  invalid graphics state
6: In doTryCatch(return(expr), name, parentenv, handler) :
  invalid graphics state

> barplot(a)
> a<-c(55,67,89,80,90)
> barplot(a)
> barplot(a,horiz=TRUE)
> a<-c(55,67,89,80,90)
> barplot(a)
> a<-c(55,67,89,80,90)
> barplot(a)
```

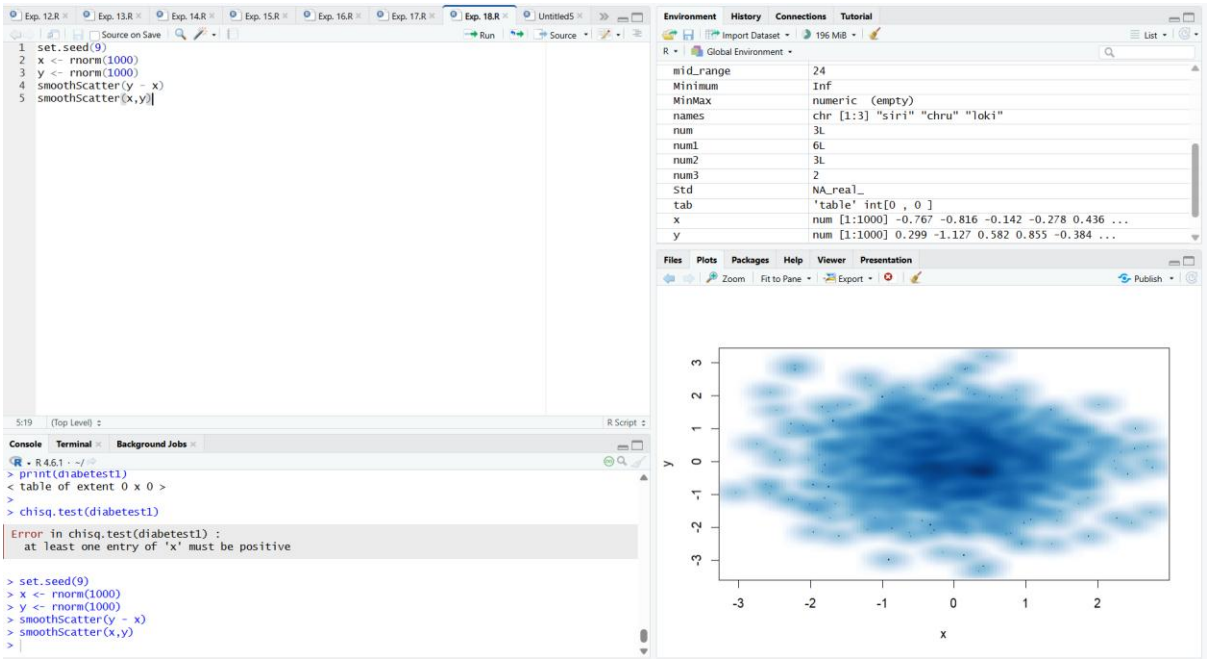
15.


names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
barplot(a)
barplot(a,horiz=TRUE)
a<-c(55,67,89,80,90)
barplot(a)
a<-c(55,67,89,80,90)
barplot(a)
barplot(a,horiz=TRUE)
names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
diabetest1 150 obs. of 5 variables
Values
a num [1:5] 55 67 89 80 90
A NULL
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
Maximum -Inf
Mean NA_real_
mid_range 24
Minimum Inf
MinMax numeric (empty)
names chr [1:3] "siri" "chru" "loki"
25.0
24.5
24.0
23.5
23.0

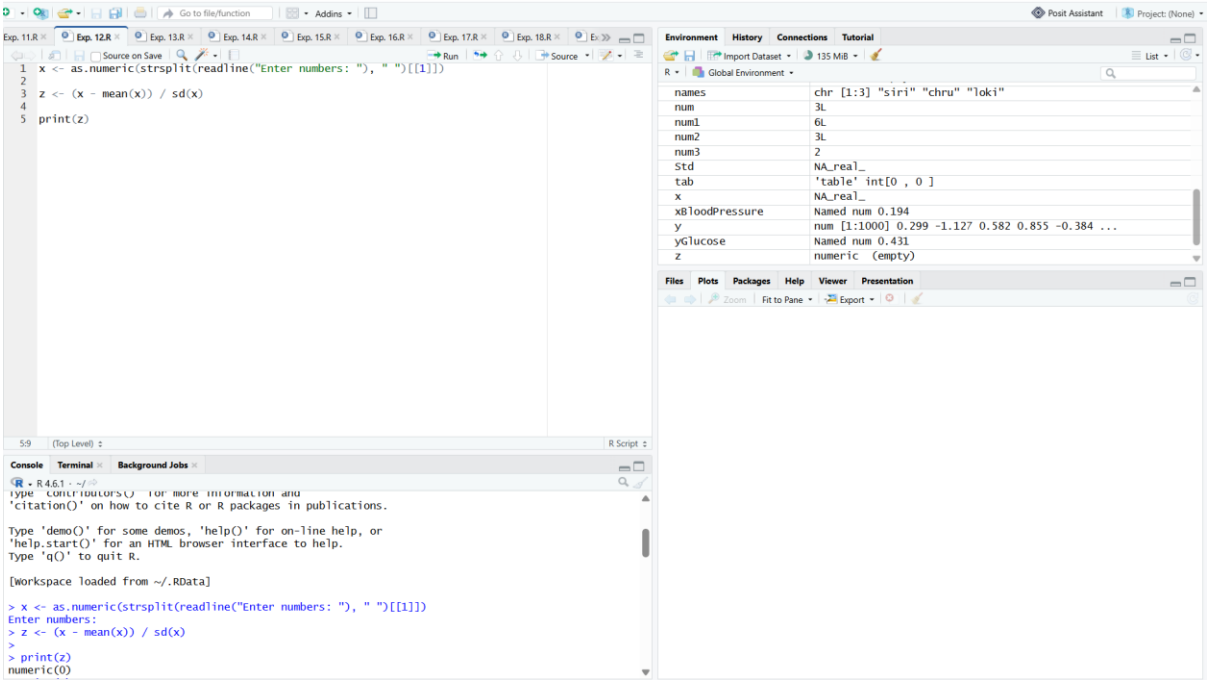
16.


a<-c(55,67,89,80,90)
hist(a)
> a<-c(55,67,89,80,90)
> barplot(a)
> a<-c(55,67,89,80,90)
> barplot(a)
> barplot(a,horiz=TRUE)
> names<-c("siri","chru","loki")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> hist(df$age)
> boxplot(df$age)
> a<-c(55,67,89,80,90)
> hist(a)
diabetest1 150 obs. of 5 variables
Values
a num [1:5] 55 67 89 80 90
A NULL
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
Maximum -Inf
Mean NA_real_
mid_range 24
Minimum Inf
MinMax numeric (empty)
names chr [1:3] "siri" "chru" "loki"
Histogram of a
Frequency
0
1
2
50
60
70
80
90
a

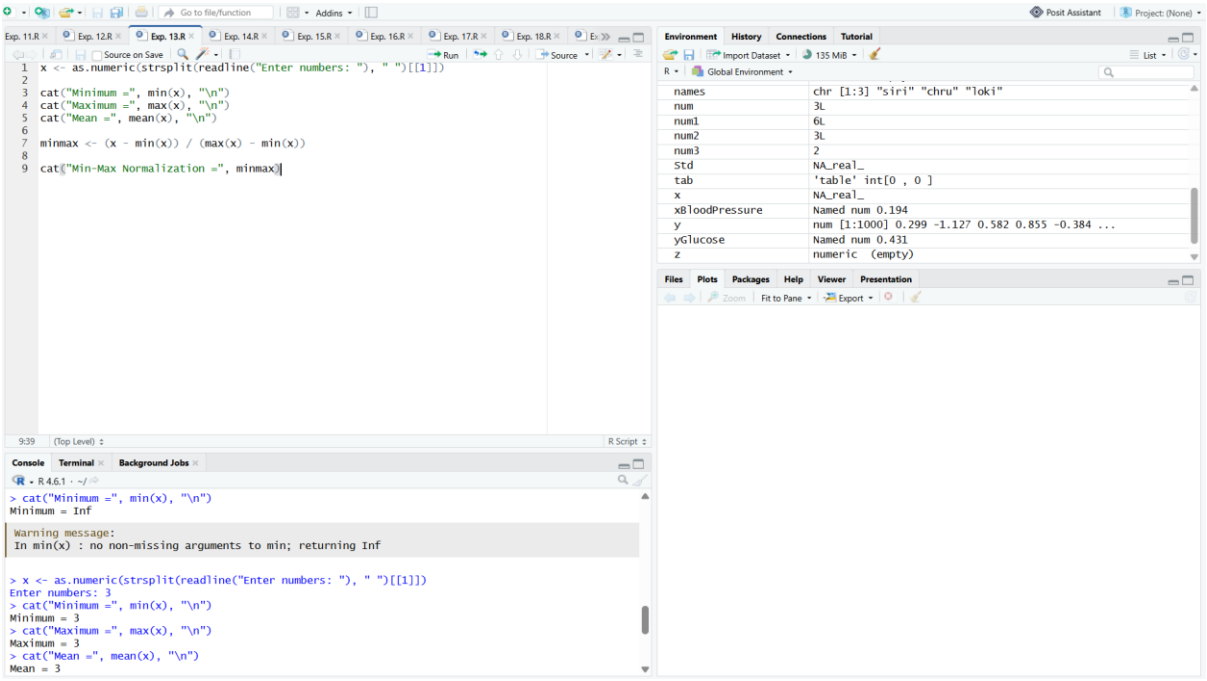
18.


set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
print(diabetest1)
< table of extent 0 x 0 >
chisq.test(diabetest1)
Error in chisq.test(diabetest1) :
at least one entry of 'x' must be positive
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
mid_range 24
Minimum Inf
MinMax numeric (empty)
names chr [1:3] "siri" "chru" "loki"
num 3L
num1 6L
num2 3L
num3 2
Std NA_real_
tab 'table' int[0 , 0]
x num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ...
y num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...
x
y

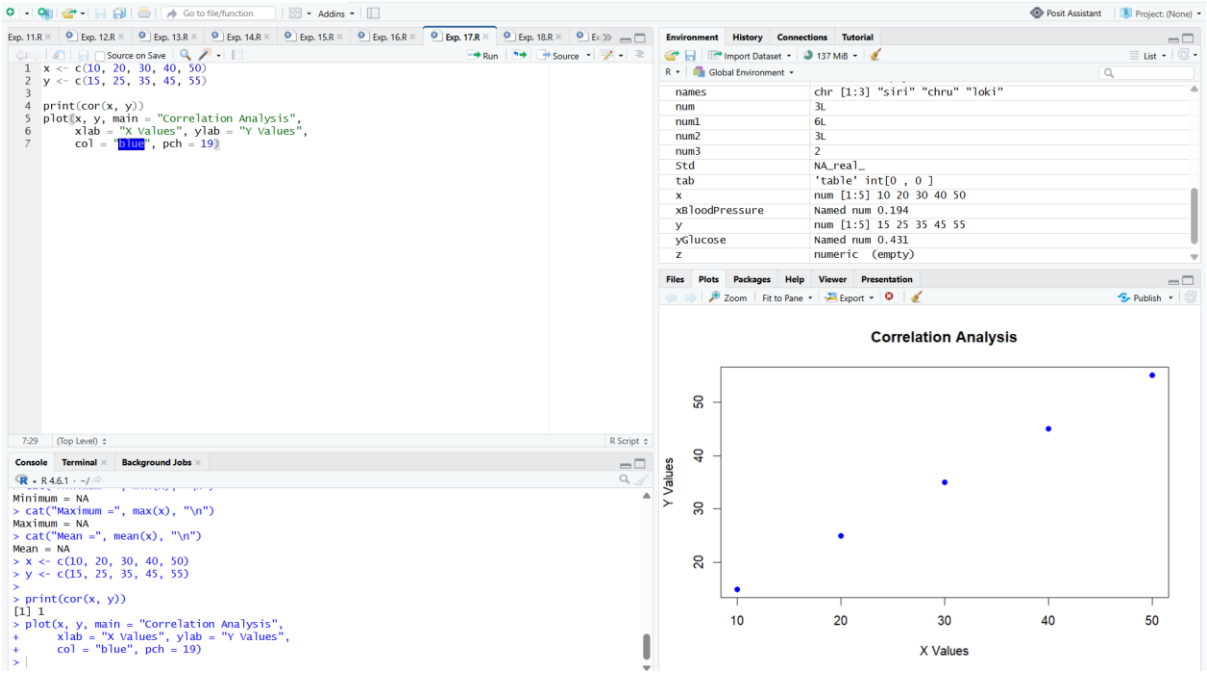
12.


x <- as.numeric(strsplit(readline("Enter numbers: "), " ")[[1]])
z <- (x - mean(x)) / sd(x)
print(z)
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> x <- as.numeric(strsplit(readline("Enter numbers: "), " ")[[1]])
Enter numbers:
> z <- (x - mean(x)) / sd(x)
> print(z)
numeric(0)
names chr [1:3] "siri" "chru" "loki"
num 3L
num1 6L
num2 3L
num3 2
Std NA_real_
tab 'table' int[0 , 0]
x NA_real_
xBloodPressure Named num 0.194
y num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...
yGlucose Named num 0.431
z numeric (empty)

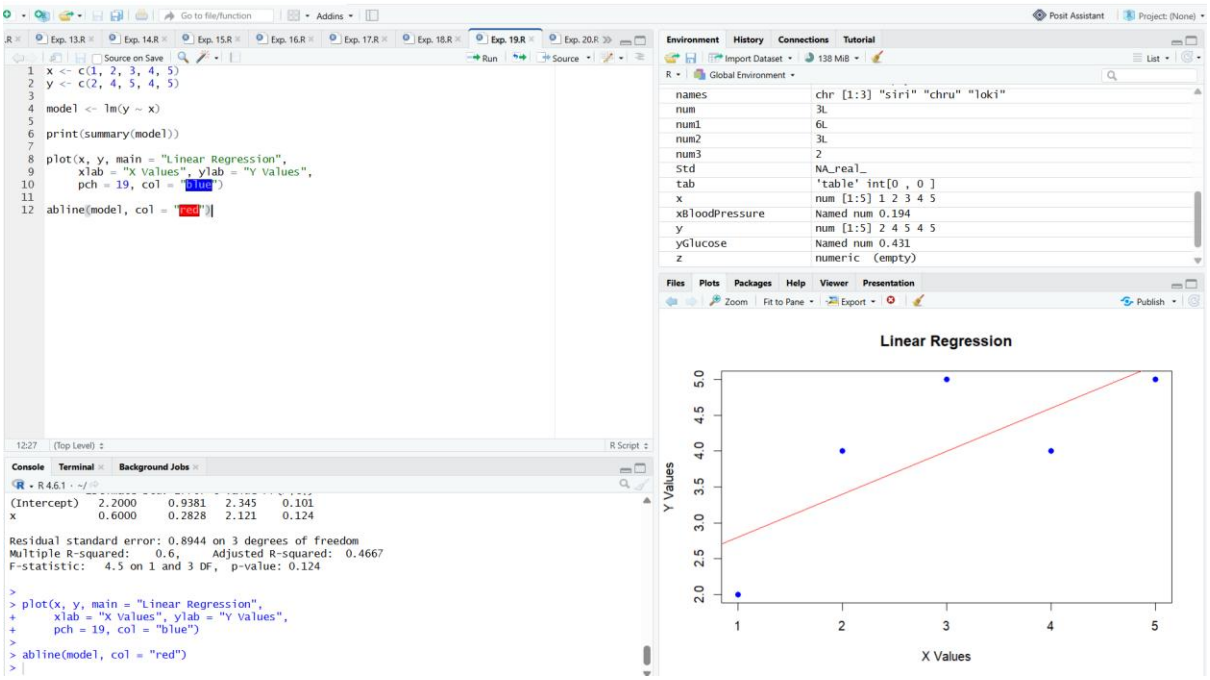
13.


x <- as.numeric(strsplit(readline("Enter numbers: "), " ")[[1]])
cat("Minimum =", min(x), "\n")
cat("Maximum =", max(x), "\n")
cat("Mean =", mean(x), "\n")
minmax <- (x - min(x)) / (max(x) - min(x))
cat("Min-Max Normalization =", minmax)
> cat("Minimum =", min(x), "\n")
Minimum = Inf
Warning message:
In min(x) : no non-missing arguments to min; returning Inf
> x <- as.numeric(strsplit(readline("Enter numbers: "), " ")[[1]])
Enter numbers: 3
> cat("Minimum =", min(x), "\n")
Minimum = 3
> cat("Maximum =", max(x), "\n")
Maximum = 3
> cat("Mean =", mean(x), "\n")
Mean = 3

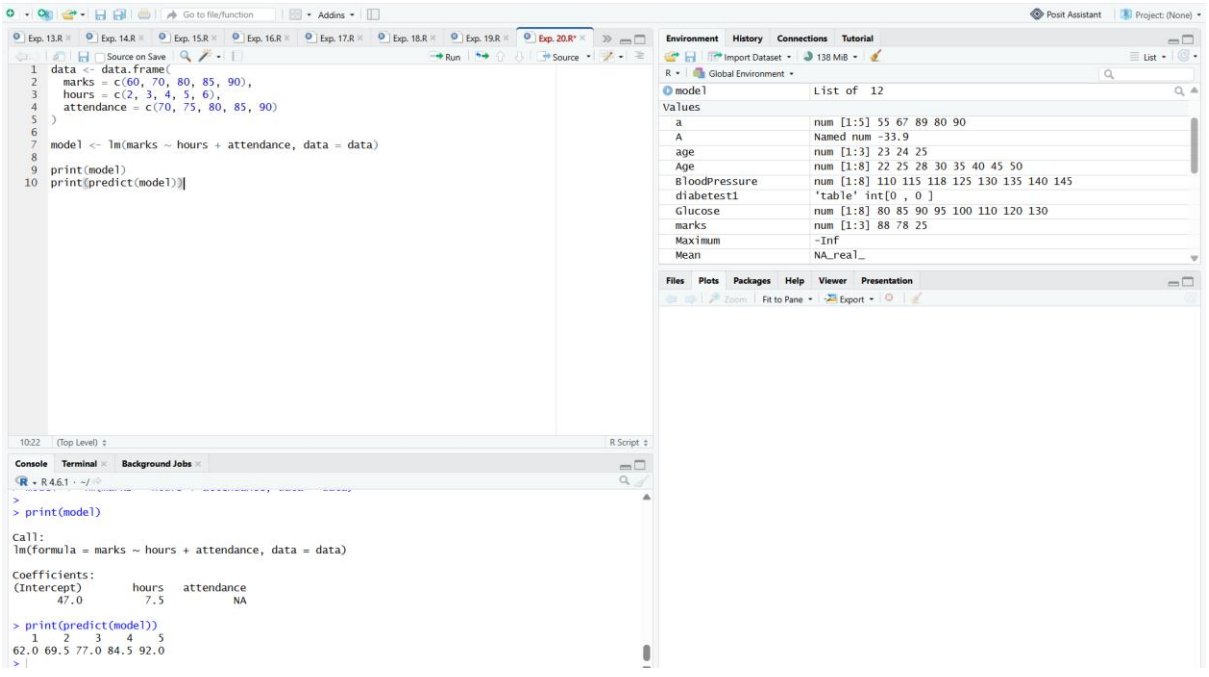
17.


x <- c(10, 20, 30, 40, 50)
y <- c(15, 25, 35, 45, 55)
print(cor(x, y))
plot(x, y, main = "Correlation Analysis",
xlab = "X Values", ylab = "Y Values",
col = "blue", pch = 19)
Correlation Analysis
Y Values
X Values
10
20
30
40
50
20
30
40
50

19.


x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 5, 4, 5)
model <- lm(y ~ x)
print(summary(model))
plot(x, y, main = "Linear Regression",
xlab = "X Values", ylab = "Y Values",
pch = 19, col = "blue")
abline(model, col = "red")
(Intercept) 2.2000 0.9381 2.345 0.101
x 0.6000 0.2828 2.121 0.124
Residual standard error: 0.8944 on 3 degrees of freedom
Multiple R-squared: 0.6, Adjusted R-squared: 0.4667
F-statistic: 4.5 on 1 and 3 DF, p-value: 0.124
Linear Regression
Y Values
X Values

20.


data <- data.frame(
marks = c(60, 70, 80, 85, 90),
hours = c(2, 3, 4, 5, 6),
attendance = c(70, 75, 80, 85, 90)
)
model <- lm(marks ~ hours + attendance, data = data)
print(model)
print(predict(model))
Call:
lm(formula = marks ~ hours + attendance, data = data)
Coefficients:
(Intercept) hours attendance
47.0 7.5 NA
1 2 3 4 5
62.0 69.5 77.0 84.5 92.0